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March 14, 2025

National Science Foundation
Networking and Information Technology Research and Development (NITRD), National Coordination Office (NCO)

Re: AI Action Plan, Request for Information

To Whom it May Concern,

The Advanced Medical Technology Association (AdvaMed) appreciates the opportunity to submit comments in response to your February 6, 2025 request for information (RFI) on the development of an artificial intelligence (AI) action plan¹.

AdvaMed is the world's largest association representing manufacturers of medical devices, diagnostic products, and medical technology. AdvaMed's member companies range from the largest to the smallest medical product innovators and manufacturers, with nearly 70 percent of our members generating less than \$100 million in annual sales. AdvaMed's member companies produce innovations that transform healthcare through earlier disease detection, less invasive procedures, and more effective treatments. AdvaMed advocates for a legal, regulatory, and economic environment that advances global healthcare by assuring worldwide patient access to the benefits of medical technology. The Association promotes policies that foster the highest ethical standards, timely product authorization, appropriate reimbursement, and access to international markets.

We recognize AI as a transformational tool with the potential to improve health outcomes, enhance efficiency of patient care, lower costs, and make advancements in healthcare. Right-sized policies can promote the development, deployment, and adoption of innovative and trustworthy AI-enabled solutions. AdvaMed is uniquely well-positioned to provide feedback on frameworks and policy considerations regarding AI because our members have been developing and deploying AI-enabled medical devices that support patient care for over 25 years.

We appreciate the opportunity to submit our high-level recommendations on priority policy actions needed to sustain and enhance AI innovation in the medical device industry.

¹ <https://www.federalregister.gov/documents/2025/02/06/2025-02305/request-for-information-on-the-development-of-an-artificial-intelligence-ai-action-plan>

Regulatory Oversight of AI-enabled Medical Devices

All medical devices, including AI-enabled devices, are subject to the Food and Drug Administration's (FDA) comprehensive and robust risk-based regulatory framework that provides for oversight of the device across the total product lifecycle. The FDA's premarket evaluation includes an assessment of device performance, reliability, and safety. After the devices are authorized for sale, device manufacturers are subject to post-market requirements including implementing robust quality management systems, device monitoring, and reporting of serious adverse events. Collectively, FDA's regulatory framework and guidance ensure that safety and performance considerations unique to AI-enabled medical devices are evaluated and implemented with the appropriate context and oversight.

As discussions about oversight of AI technologies across industries continue to evolve, it is crucial to distinguish between the different applications of AI (e.g., medical devices vs. self-driving cars) and ensure that regulations are sector-specific and appropriately tailored to the respective uses and needs. Legislation or regulation that attempts to address AI across all industries has the potential to introduce redundant or conflict requirements with existing medical device regulations, slowing innovation and delaying timely patient access to important medical devices. Overly broad and burdensome regulations risk stifling innovation, diminishing patient care and adversely impacting global competitiveness.

FDA should remain the lead regulator responsible for overseeing the safety and effectiveness of AI-enabled medical devices. The existing FDA regulatory framework remains well suited for AI-enabled medical devices and FDA has the best understanding of the unique considerations and risks specific to AI-enabled medical devices. We encourage the U.S. government to continue to leverage FDA's existing capabilities and regulatory framework to avoid duplication or contradiction in regulatory approaches.

FDA Resources

While public discussion regarding AI has increased in recent years, the utilization of AI is not new for the medical device industry, which has been developing and deploying FDA-regulated AI-enabled medical devices for more than 25 years². As of December 2024, the FDA has authorized over one thousand AI-enabled devices across a variety of medical specialties including radiology, oncology, cardiology, and neurology. This number is only expected to grow, especially as the number of clinical applications expands and their clinical importance continues to be demonstrated. Therefore, it is critically important to safeguard the process of getting the latest and most promising medical technologies into the hands of U.S. doctors and patients. An effective FDA is an essential element of that process. Insufficient staffing and skills at FDA will slow-down product review times, reducing the ability of America to be first in the world to benefit from the most innovative products, including AI-enabled medical devices. An efficient, transparent, and effective FDA is enabled through user fee agreements that ensure it is adequately staffed with the technical expertise necessary to keep pace with innovation.

² <https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-aiml-enabled-medical-devices>

Performance Assurance & Safety Standards

In recent years, healthcare stakeholders have begun exploring the creation of third-party quality assurance labs focused on performance monitoring. However, we have concerns about the suitability and practicality of this approach for regulated medical devices. Given FDA's existing oversight of AI-enabled medical devices, third-party labs for regulated medical devices are redundant. If third-party labs were linked to the regulatory process, their use would effectively become mandatory, increasing costs and burdens for manufacturers who already evaluate their devices in accordance with best practices like those cataloged in FDA-recognized international consensus standards.

We are also concerned that third-party evaluators, such as assurance labs, may utilize proprietary methods and metrics that would lack transparency to developers, patients, and providers. There are also significant concerns about their handling of sensitive training and testing data. Mandating manufacturers to share proprietary information about AI-enabled devices with external labs raises confidentiality, intellectual property (IP), and security issues as variability in lab security practices for third-party labs could expose or jeopardize sensitive manufacturer or patient information. Implementing a third-party framework of labs creates, at best, redundancy with existing FDA oversight while potentially increasing costs, slowing innovation, and limiting access to valuable medical devices, without clear benefit to clinicians or patients.

Rather than create a new framework of third-party evaluators, policymakers should continue to encourage FDA to participate in the development and timely recognition of accredited and consensus-based standards for safety and quality assurance processes. The longstanding practice of relying on collaboratively developed, international consensus standards promotes a deeper understanding of both the benefits and risks of these technologies while improving transparency. Enabling manufacturers to utilize consensus standards allows for robust quality and safety assurance that maintains data security and IP confidentiality. This approach supports patient safety and innovation by minimizing the need for external assessments, which could introduce added costs, regulatory delays, and potential security vulnerabilities.

International Collaboration

AdvaMed members support a globally harmonized regulatory approach for AI-enabled medical devices. We support the recent collaborative efforts between FDA, Health Canada, and UK MHRA to establish a common perspective on good machine learning (ML) practices for medical device development³, PCCP (Predetermined Change Control Plans) for ML-enabled medical devices⁴, and transparency for ML-

³ <https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles>

⁴ <https://www.fda.gov/medical-devices/software-medical-device-samd/predetermined-change-control-plans-machine-learning-enabled-medical-devices-guiding-principles>

enabled medical devices⁵. We encourage FDA and policymakers to continue to consider policies and practices that support global alignment, where practical.

FDA must ensure the tools and processes it has under its existing authority are implemented and, where appropriate, adapted to keep pace with emerging technologies. As AI technology matures, international consensus standards specific to medical devices should be the foundation for safe, effective, and responsible AI model development and deployment. It is critical that FDA prioritize the development, revision, and timely recognition of such standards to promote industry-wide adoption of these best practices. Prioritizing international consensus standards would support consistency in FDA's expectations for AI-enabled medical technologies while harmonizing with globally recognized best practices.

Payment/Reimbursement

Appropriate reimbursement for the adoption and use of AI-enabled medical technologies is critical to ensuring patients have timely access to and benefit from these innovations. As the nation's largest payer of health care, Medicare's policies on coverage and payment for AI-enabled medical technologies are especially critical, applying to the millions-strong Medicare population, and because private payers and state Medicaid plans often look to Medicare as they establish their own coverage policies.

We believe Medicare has regulatory authority to expand access to AI-enabled medical technologies. However, its regulatory framework currently lacks the specificity and clarity to provide coverage and payment for digital health technologies broadly and for AI-enabled medical technologies and software specifically. The result has been incremental, technology-specific policy changes, with many AI and software innovators struggling to find pathways to coverage and reimbursement for new technologies.

There is no "one size fits all" reimbursement policy for every AI-enabled medical technology. Instead, appropriate payment mechanisms vary depending on the kind of technology in question and the clinical setting in which it is used. Regardless, accurately capturing the cost and value of these technologies is critical to ensuring appropriate reimbursement.

American patients deserve access to safe, proven effective medical technology that can improve their health outcomes and meet underserved clinical needs, such as mental health services. Patients stand to gain greatly from the development and adoption of digital health and AI-enabled medical technologies that improve the diagnosis and treatment of illness and disability, promote healthy behaviors, and support population health management. Appropriate reimbursement policies will enable doctors, hospitals, and other caregivers to adopt AI-enabled medical technologies into their practices, improving the care they can give patients.

Data Access & Data Privacy

AI is distinguished from other health technologies by its ability to analyze vast datasets to assist physicians in diagnosis and treatment, supplement and enhance the clinical process, and offer

⁵ <https://www.fda.gov/medical-devices/software-medical-device-samd/transparency-machine-learning-enabled-medical-devices-guiding-principles>

personalized health care solutions. AI can provide insights that may not otherwise be available via conventional health care technologies or techniques and rapidly process data to produce clinical support and recommendations that can be used to inform care decisions.

Large, diverse data sets are needed by AI medical device developers to train and validate trustworthy algorithms. Unlocking the potential of AI-driven health care solutions is linked to the availability of high-quality data to build and evaluate these technologies. Challenges such as the fragmented nature of health care data, non-standardized data formats, difficulty accessing data across different health systems, and the lack of interoperability between platforms impede the pace of innovation. Flexibility to pursue different approaches to obtaining and utilizing data is crucial to ensuring innovation is not limited by data access.

Data aggregation and access processes are currently siloed and complex to navigate. Data needed to produce valuable health care insights is spread across many different data aggregators and third-party data vendors whose data is not standardized, leading to disparate data quality and utility. Additionally, there are only a handful of commercial vendors that provide services needed to link data (e.g., for purposes of tokenization and expert determination).

In undertaking data collection, it is crucial to address the data protection requirements for the large-scale processing of health data that powers AI models. Safeguarding patient privacy and ensuring robust data security are vital to protecting sensitive health information. Moreover, informed notice and patient autonomy are important to meaningful patient involvement in AI-driven health care decisions.

While AdvaMed supports robust privacy and security protections for patient data, current federal laws pose difficulties for AI developers to access and utilize large-scale datasets with intact metadata which are needed to provide solutions for patients. The Health Insurance Portability and Accountability Act of 1996 (HIPAA) required the creation of national standards to protect sensitive patient health information from being disclosed without the patient's consent or knowledge. HIPAA and its implementing regulations impose strict requirements on the collection, storage, use, and disclosure of patient health data resulting in challenges for developers to access sufficient training data, especially when it has undergone anonymization.

Data quality and provenance are important considerations for AI medical device developers in the training and validation of AI models. Privacy law requirements for de-identification and/or minimization of personal data or metadata can be inconsistent and at tension with these important considerations. These restrictions may impact the ability of AI medical device developers to:

- access, store and retain training and validation datasets (and metadata) over a certain period of time to meet FDA's recommendations; and
- demonstrate that the dataset used to train and tune the device is robust and representative of the intended patient population. To conduct a bias analysis, for example, patient demographic and health information may be required (e.g., ethnicity, sex, gender, age, and any relevant clinical

indications). This information can be hard to obtain, or it may be difficult to negotiate retention periods or data use rights.

As such, we respectfully request that the National Science Foundation consider the following when drafting an artificial intelligence (AI) action plan:

- ***Ensure data protection without stifling innovation.*** Develop a pragmatic approach for AI in health care that promotes the development of privacy-preserving techniques, balancing innovation and the development of AI solutions with the need to protect sensitive health data.
 - Advance comprehensive federal legislation with strong preemption that differentiates health data from general consumer data due to its unique role in patient safety, care coordination, and innovation.
- ***Evaluate updating HIPAA for the AI era and provide clear guidance for health data use in AI development.*** Ensure that HIPAA standards allow for the sharing of the datasets needed to train, test, validate, and re-train AI models while preserving patient privacy. The current HIPAA de-identification methods (authorization, safe harbor, and expert determination) stifle the high-volume data usage and sharing that can optimize the development of safe and accurate AI models.
- ***Develop appropriate guidelines around patient notice and authorization for the data used to develop AI.*** More data will allow for the creation of better, more accurate AI models, ultimately leading to better outcomes for patients. Patient notice and authorization should be at the center of additional data accessibility. New frameworks for health data regulation should consider any data limitations presented by HIPAA and other privacy laws.

We recognize that the pace of innovation is fast. As federal legislators seek to ensure AI-enabled products in all industries are used safely, we appreciate the opportunity to provide feedback on policy priorities and recommendations related to medical devices. AdvaMed member companies take seriously the level of trust placed in them by patients and have consistently taken action to self-identify best practices that balance innovation with patient protections. Thank you for the opportunity to submit these comments. Please consider AdvaMed as a resource on med-tech regulatory, data stewardship, reimbursement, and privacy matters as you consider policies related to AI and medical devices.

Statement provided in accordance with the RFI Instructions: This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Respectfully Submitted,

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